

Solving Linear Programming Problems

The Simplex Method (PART 2)

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Outline

Key Takeaways from Last Lecture

Questions Regarding Simplex

Two-Phase method

Simplex: Special cases

- Infeasibility

- Unboundedness

- Alternative Optima

- Degeneracy and cycling

Sensitivity Analysis

Key Takeaways from Last Lecture

- Simplex is an efficient algorithm for finding optimal solutions to LP problems by navigating through the corner points of the feasible region.
- It iteratively moves from one Basic Feasible Solution (BFS) to a better neighborhood BFS until the optimal BFS is reached.
- By detecting the optimal BFS, the simplex method provides the optimal values of the decision variables and the objective function.

Questions Regarding Simplex

- How can we choose an appropriate initial BFS if the origin is not a basic feasible solution (BFS)?

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- How can we choose an appropriate initial BFS if the origin is not a basic feasible solution (BFS)?
- What are the special cases that may arise when using Simplex?
- Does Simplex terminate in every LP?

Artificial starting solution

- LPs where all constraints are of the form " $\leq$ " with nonnegative right-hand sides can be conveniently started with an all-slack basic feasible solution.
- Select "all-slack" variables as basic variables to create an initial basic feasible solution (BFS).

$$\begin{aligned} s_1 &= b_1 + \sum_{i=m+1}^n a_{1i}x_i \\ &\vdots \\ s_m &= b_m + \sum_{i=m+1}^n a_{mi}x_i \end{aligned}$$

Artificial starting solution

- If LPs involves constraints of the form " $\geq$ " or " $=$ " do not have this convenient starting solution.
- This "ill-behaved" LPs, artificial variables should be used to find an initial BFS that we can start with.

$$\begin{array}{ll}\text{Maximize} & 3x + 9y \\ \text{Subject to} & x + y \leq 3 \\ & 5x - y \geq 3 \\ & y \geq 1\end{array}$$

Two-Phase method

$$\text{Maximize } c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

$$\text{Subject to } a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

$$x_1, x_2, \dots, x_n \geq 0$$

Two-Phase method

Minimize $R_1 + R_2 + \cdots + R_m$

Subject to $a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n + \mathbf{R}_1 = b_1$

$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n + \mathbf{R}_2 = b_2$

$\vdots$

$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n + \mathbf{R}_m = b_m$

- Feasible if the objective value reaches 0.
- All the $\mathbf{R}_i$ are zeros.
- I have a BFS without the R_i .

Example: The Standard form (Phase 1)

$$\text{Maximize } z = 3x + 9y$$

$$\text{Subject to } x + y \leq 3$$

$$5x - y \geq 3$$

$$y \geq 1$$

Example: The Standard form (Phase 1)

$$\text{Maximize } z = 3x + 9y$$

$$\text{Subject to } x + y + s_1 = 3$$

$$5x - y - s_2 = 3$$

$$y - s_3 = 1$$

Example: Two-Phase method (Phase 1)

$$\begin{array}{ll}\text{Minimize} & z = R_1 + R_2 + R_3 \\ \text{Subject to} & x + y + S_1 + R_1 = 3 \\ & 5x - y - S_2 + R_2 = 3 \\ & y - S_3 + R_3 = 1\end{array}$$

Example: Two-Phase method (Phase 1)

Basic	x	y	s_1	s_2	s_3	R_1	R_2	R_3	RHS	Ratio
z	0	0	0	0	0	-1	-1	-1	0	
R_1	1	1	1	0	0	1	0	0	3	
R_2	5	-1	0	-1	0	0	1	0	3	
R_3	0	1	0	0	-1	0	0	1	1	

- Basic variables = $\{R_1, R_2, R_3\}$
- We should remove basic variables from the objective function to start Simplex.
- $\text{Row}(z) = \text{Row}(z) + \text{Row}(R_1) + \text{Row}(R_2) + \text{Row}(R_3)$

Example: Two-Phase method (Phase 1)

Basic	x	y	S_1	S_2	S_3	R_1	R_2	R_3	RHS	Ratio
z	6	1	1	-1	-1	0	0	0	7	
R_1	1	1	1	0	0	1	0	0	3	
R_2	5	-1	0	-1	0	0	1	0	3	
R_3	0	1	0	0	-1	0	0	1	1	

- Entering variable: S_1 Leaving Variable: R_1
- Basic variables = $\{S_1, R_2, R_3\}$
- We should remove basic variables from the objective function to start Simplex.
- $\text{Row}(z) = \text{Row}(z) - \text{Pivot row}$

Example: Two-Phase method (Phase 1)

Basic	x	y	S_1	S_2	S_3	R_1	R_2	R_3	RHS	Ratio
z	5	0	0	-1	-1	-1	0	0	4	
S_1	1	1	1	0	0	1	0	0	3	3
R_2	5	-1	0	-1	0	0	1	0	3	3/5
R_3	0	1	0	0	-1	0	0	1	1	

- Entering variable: x , Leaving Variable: R_2
- Basic variables = $\{S_1, x, R_3\}$
- $\text{Row}(z) = \text{Row}(z) - 5 \cdot \text{Pivot row}$
- $\text{Row}(S_1) = \text{Row}(S_1) - \text{Pivot row}$

Example: Two-Phase method (Phase 1)

Basic	x	y	S_1	S_2	S_3	R_1	R_2	R_3	RHS	Ratio
z	0	1	0	0	-1	-1	-1	0	1	
S_1	0	6/5	1	1/5	0	1	-1/5	0	12/5	2
x	1	-1/5	0	-1/5	0	0	1/5	0	3/5	
R_3	0	1	0	0	-1	0	0	1	1	1

- Entering variable: y , Leaving Variable: R_3
- Basic variables = $\{S_1, x, y\}$
- $\text{Row}(z) = \text{Row}(z) - \text{Pivot row}$
- $\text{Row}(S_1) = \text{Row}(S_1) - \frac{6}{5} \text{Pivot row}$
- $\text{Row}(x) = \text{Row}(x) + \frac{1}{5} \text{Pivot row}$

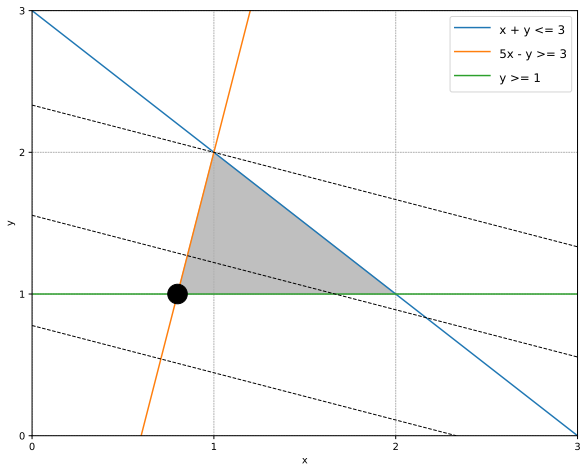
Example: Two-Phase method (Phase 1)

Basic	x	y	S_1	S_2	S_3	R_1	R_2	R_3	RHS	Ratio
z	0	0	0	0	0	-1	-1	-1	0	
S_1	0	0	1	1/5	6/5	1	-1/5	-6/5	6/5	
x	1	0	0	-1/5	-1/5	0	1/5	1/5	4/5	
y	0	1	0	0	-1	0	0	1	1	

- Optimally detected because $(C_i \leq 0, \forall i)$.
- $R_1 = R_2 = R_3 = 0$
- Basic variables = $\{S_1, x, y\}$
- Non-basic variables = $\{S_2, S_3\}$

Example: Two-Phase method (Phase 2)

$$\begin{array}{llllll} \text{Maximize} & z & = & 3x & + & 9y \\ \text{Subject to} & s_1 & = & \frac{6}{5} & + & \frac{1}{5}s_2 & + & \frac{6}{5}s_3 \\ & x & = & \frac{4}{5} & - & \frac{1}{5}s_2 & - & \frac{1}{5}s_3 \\ & y & = & 1 & & & - & s_3 \end{array}$$



Example: Two-Phase method (Phase 2)

Basic	x	y	s_1	s_2	s_3	RHS	Ratio
z	-3	-9	0	0	0	0	
s_1	0	0	1	1/5	6/5	6/5	
x	1	0	0	-1/5	-1/5	4/5	
y	0	1	0	0	-1	1	

- Basic variables = $\{s_1, x, y\}$
- We should remove basic variables from the objective function to start Simplex.
- $\text{Row}(z) = \text{Row}(z) + 3 \times \text{Row}(x) + 9 \times \text{Row}(y)$

Example: Two-Phase method (Phase 2)

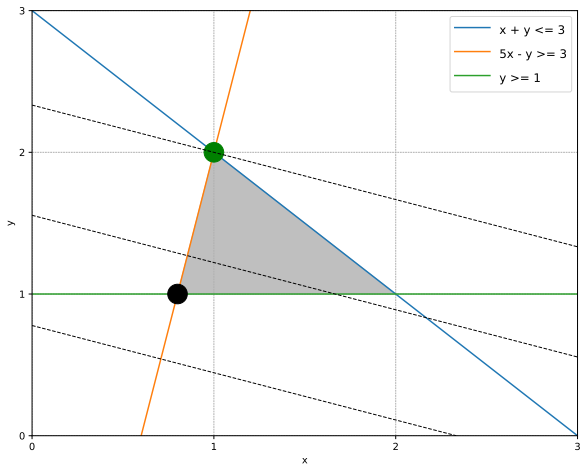
Basic	x	y	s_1	s_2	s_3	RHS	Ratio
z	0.00	0.00	0.00	-3/5	-48/5	57/5	
s_1	0.00	0.00	1.00	1/5	6/5	6/5	
x	1	0	0	-1/5	-1/5	4/5	
y	0	1	0	0	-1	1	

- Entering variable: s_3 , Leaving Variable: s_1
- **Pivot row** = $\frac{5}{6} \times$ **Pivot row**
- **Row(z)** = **Row(z)** + $\frac{48}{5} \times$ **Pivot row**
- **Row(x)** = **Row(x)** + $\frac{1}{5} \times$ **Pivot row**
- **Row(y)** = **Row(y)** + **Pivot row**

Example: Two-Phase method (Phase 2)

Basic	x	y	s_1	s_2	s_3	RHS	Ratio
z	0.00	0.00	8.00	1.00	0.00	21.00	
s_3	0	0	5/6	1/6	1	1	
x	1	0	1/6	-1/6	0	1	
y	0	1	5/6	1/6	0	2	

- Optimally detected because $(\mathbf{c}_i \geq \mathbf{0}, \forall i)$.
- Basic variables = $\{s_3, x, y\}$
- Non-basic variables = $\{s_1, s_2\}$
- Optimum $(\mathbf{x} = 1, \mathbf{y} = 2)$ and $\mathbf{z} = 21$.



Simplex: Special cases

- **Infeasibility:** occurs when there is no feasible solution that satisfies all of the constraints.
- **Unboundedness:** occurs when the objective function can be increased indefinitely without violating any of the constraints.
- **Alternative Optima:** occurs when several global optima with same objective value exists.
- **Degeneracy:** occurs when one or more basic variables become zero during the iteration process.

Special cases: Infeasibility

- Empty feasible region.
- Can be detected using Two-Phase method.
- The objective function ($\sum R_i$) in Phase 1 cannot be 0.

$$\begin{array}{ll} \text{Maximize} & z = 2x_1 - 3x_2 \\ \text{S.t.} & x_1 + x_2 \leq 2 \\ & 2x_1 - 2x_2 \geq 5 \\ & x_1, x_2 \geq 0 \end{array}$$

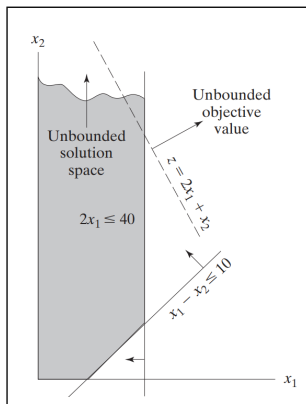
Special cases: Unboundedness

- Unbounded solutions allow for arbitrary increases in variables without violating any constraints.
- Unboundedness may indicate a poorly constructed model.

$$\begin{array}{llllll} \textbf{Maximize} & z = & 2 & x_1 & + & x_2 \\ \textbf{S.t} & & & x_1 & - & x_2 \leq 10 \\ & & 2 & x_1 & & \leq 40 \\ & & & x_1 & , & x_2 \geq 0 \end{array}$$

Special cases: Unboundedness

$$\begin{array}{ll} \text{Maximize} & z = 2x_1 + x_2 \\ \text{s.t} & x_1 - x_2 \leq 10 \\ & 2x_1 \leq 40 \\ & x_1, x_2 \geq 0 \end{array}$$



How to detect Unboundedness

Simplex Method indicates unbounded solutions when all Ratios values are either infinite or negative, resulting in no leaving variable.

Basic	x_1	x_2	s_1	s_2	RHS	Ratio
z	-2	-1	0	0	0	
s_1	1	-1	1	0	10	Negative
s_2	2	0	0	1	40	Infinite

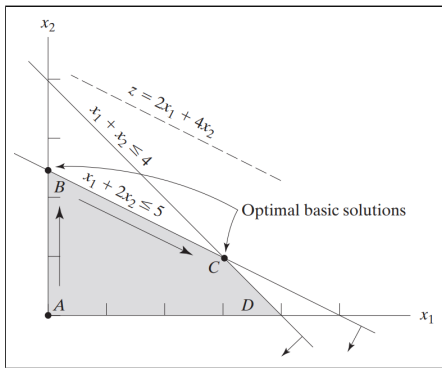
Special cases: Alternative Optima

- An LP problem may have infinite alternative optima when the objective function is parallel to a constraint.
- Any point on that constraint line is also optimal.
- Alternative optima provide different variable combinations with the same optimal objective value.

$$\begin{array}{ll}\text{Maximize} & z = 2x_1 + 4x_2 \\ \text{S.t} & x_1 + 2x_2 \leq 5 \\ & x_1 + x_2 \leq 4 \\ & x_1, x_2 \geq 0\end{array}$$

Special cases: Alternative Optima

$$\begin{array}{ll} \text{Max} & z = 2x_1 + 4x_2 \\ \text{s.t} & x_1 + 2x_2 \leq 5 \\ & x_1 + x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{array}$$



Special cases: Alternative Optima

Basic	x_1	x_2	s_1	s_2	RHS
z	-2	-4	0	0	0
s_1	1	2	1	0	5
s_2	1	1	0	1	4

- Entering variable: x_2 , Leaving Variable: s_1
- **Pivot row** = $\frac{1}{2} \times$ **Pivot row**
- **Row(z)** = **Row(z)** + 4 $\times$ **Pivot row**
- **Row(s_2)** = **Row(s_2)** - **Pivot row**

Special cases: Alternative Optima

- If a non-basic variable **has a zero coefficient**, it can be replaced with a basic variable whose right-hand side value is strictly positive without changing the objective function's right-hand side value.
- If we swap x_1 with x_2 , where x_2 has a right-hand side value of 2.5, then the objective function's right-hand side value remains at 10. After the swap, x_1 becomes the new basic variable with a value of 5, and x_2 becomes a non-basic variable with a value of 0.

Basic	x_1	x_2	s_1	s_2	RHS
z	0	0	2	0	10
x_2	0.5	1	0.5	0	2.5
s_2	0.5	0	-0.5	1	1.5

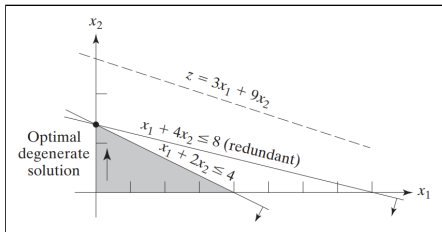
Special cases: Degeneracy

- Feasibility condition of simplex method can have ties for minimum ratio.
- Ties can be broken arbitrarily but will result in a degenerate solution in the next iteration
- Degeneracy can cause the algorithm to cycle indefinitely and not terminate

$$\begin{array}{ll}\text{Maximize} & z = 3x_1 + 9x_2 \\ \text{S.t} & x_1 + 4x_2 \leq 8 \\ & x_1 + 2x_2 \leq 4 \\ & x_1, x_2 \geq 0\end{array}$$

Special cases: Degeneracy

$$\begin{array}{lll} \text{Max} & z = 3x_1 + 9x_2 & \\ \text{s.t} & x_1 + 4x_2 \leq 8 & \\ & x_1 + 2x_2 \leq 4 & \\ & x_1, x_2 \geq 0 & \end{array}$$



Special cases: Degeneracy

- Ties in minimum ratio.
- Some basic variable are equal to zero.

Basic	x_1	x_2	s_1	s_2	RHS	Ratio
z	-3	-9	0	0	0	
s_1	1	4	1	0	8	2
s_2	1	2	0	1	4	2

Special cases: Degeneracy

- Ties in minimum ratio.
- Some basic variable are equal to zero.

Basic	x_1	x_2	s_1	s_2	RHS	Ratio
z	-0.75	0	2.25	0	18	
x_2	0.25	1	0.25	0	2	8
s_2	0.5	0	-0.5	1	0	0

Special cases: Degeneracy

- Ties in minimum ratio.
- Some basic variable are equal to zero.

Basic	x_1	x_2	s_1	s_2	RHS	Ratio
z	0	0	1.5	1.5	18	
x_2	0	1	0.5	-0.5	2	
x_1	1	0	-1	2	0	

Degeneracy interpretation

- The presence of degeneracy in an LP suggests the potential existence of a superfluous constraint.
- Shuffling around the basic variables without departing from a corner.
- Dealing with degeneracy in an LP can create the impression that we are moving from one corner to another, while keeping the objective value constant.

Degeneracy can cause cycling

- Cycling happens when the simplex algorithm loops between multiple solutions without reaching the optimal solution due to degeneracy.
- This can cause the simplex algorithm to loop indefinitely.
- To prevent cycling, anti-cycling rules, such as Bland's rule, can be applied to stop revisiting the same solution and improve the efficiency of the simplex algorithm.
- If there are multiple ratios that are minimal, choose the variable x_j with the smallest index as the entering variable.

Example of cycling

$$\begin{array}{ll}
 \text{Max} & z = 2.3x_1 + 2.15x_2 - 13.55x_3 - 0.4x_4 \\
 \text{s.t} & 0.4x_1 + 0.2x_2 - 1.4x_3 - 0.2x_4 \leq 0 \\
 & -7.8x_1 - 1.4x_2 + 7.8x_3 + 0.4x_4 \leq 0 \\
 & x_1, x_2, x_3, x_4 \geq 0
 \end{array}$$

Example of cycling

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	-2.3	-2.15	13.55	0.4	0	0	0
x_5	0.4	0.2	-1.4	-0.2	1	0	0
x_6	-7.8	-1.4	7.8	0.4	0	1	0

We are at the origin $(0, 0, \dots, 0)$

Example of cycling

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	0	-1	5.5	-0.75	5.75	0	0
x_1	1	0.5	-3.5	-0.5	2.5	0	0
x_6	0	2.5	-19.5	-3.5	19.5	1	0

We are at the origin $(0, 0, \dots, 0)$

Example of cycling

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	0	0	-2.3	-2.15	13.55	0.4	0
x_1	1	0	0.4	0.2	-1.4	-0.2	0
x_2	0	1	-7.8	-1.4	7.8	0.4	0

We are at the origin $(0, 0, \dots, 0)$.

Example of cycling

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	5.75	0	0	-1	5.5	-0.75	0
x_3	2.5	0	1	0.5	-3.5	-0.5	0
x_6	19.5	1	0	2.5	-19.5	-3.5	0

We are at the origin $(0, 0, \dots, 0)$.

Example of cycling

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	13.55	0.4	0	0	-2.3	-2.15	0
x_3	-1.4	-0.2	1	0	0.4	0.2	0
x_4	7.8	0.4	0	1	-7.8	-1.4	0

We are at the origin $(0, 0, \dots, 0)$.

Example of cycling

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	5.5	-0.75	5.75	0	0	-1	0
x_5	-3.5	-0.5	2.5	0	1	0.5	0
x_4	-19.5	-3.5	19.5	1	0	2.5	0

We are at the origin $(0, 0, \dots, 0)$.

Example of cycling

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	-2.3	-2.15	13.55	0.4	0	0	0
x_5	0.4	0.2	-1.4	-0.2	1	0	0
x_4	-7.8	-1.4	7.8	0.4	0	1	0

The Simplex has returned to its original state.

Example of cycling

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	-2.3	-2.15	13.55	0.4	0	0	0
x_5	0.4	0.2	-1.4	-0.2	1	0	0
x_6	-7.8	-1.4	7.8	0.4	0	1	0

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	-2.3	-2.15	13.55	0.4	0	0	0
x_5	0.4	0.2	-1.4	-0.2	1	0	0
x_4	-7.8	-1.4	7.8	0.4	0	1	0

The Simplex will continuously cycle through these states.

Sensitivity Analysis

- **Sensitivity analysis** (or post-optimality analysis) determines how optimal solutions are affected by changes within specified ranges.
 - Changes in **right-hand side (RHS) values**.
 - Changes in **objective function coefficients**.

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- Managers must operate in dynamic environments with **imprecise estimates** of coefficients.

Sensitivity Analysis

- **Sensitivity analysis** (or post-optimality analysis) determines how optimal solutions are affected by changes within specified ranges.
 - Changes in **right-hand side (RHS) values**.
 - Changes in **objective function coefficients**.
- Managers must operate in dynamic environments with **imprecise estimates** of coefficients.
- Sensitivity analysis is important for managers to ask **“what-if”** questions about the problem.

Graphical sensitivity Analysis

- We consider two cases:
 1. Sensitivity of the optimum solution to changes in the availability of the resources (right-hand side of the constraints)
 2. Sensitivity of the optimum solution to changes in unit profit or unit cost (coefficients of the objective function)

Changes in the Right-Hand side

- JOBCO manufactures two products on two machines.
- Processing times and revenues per unit are given as follows:
 - **Product 1:** 2 hrs on machine 1, 1 hr on machine 2, \$30 revenue per unit.

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Changes in the Right-Hand side

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- Processing times and revenues per unit are given as follows:
 - **Product 1:** 2 hrs on machine 1, 1 hr on machine 2, \$30 revenue per unit.
 - **Product 2:** 1 hr on machine 1, 3 hrs on machine 2, \$20 revenue per unit
- Total daily processing time available for each machine is 8 hrs
- x_1 and x_2 represent the daily number of units of products 1 and 2.

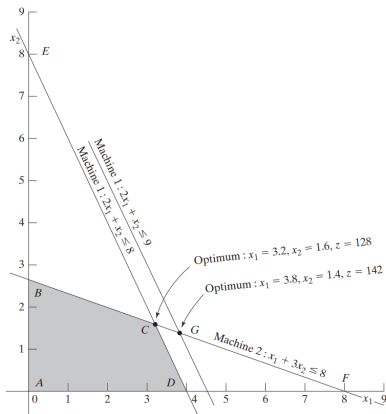
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- Total daily processing time available for each machine is 8 hrs
- x_1 and x_2 represent the daily number of units of products 1 and 2.

$$\begin{array}{llllllll} \text{Maximize} & z = & 30 & x_1 & + & 20 & x_2 & \\ \text{S.t} & & 2 & x_1 & + & & x_2 & \leq 8 \\ & & & x_1 & + & 3 & x_2 & \leq 8 \\ & & & x_1 & , & & x_2 & \geq 0 \end{array}$$

Changes in the Right-Hand side

- Increasing machine 1 capacity from 8 to 9 hrs moves the optimum solution to point G.
- $$\frac{\text{Rate of revenue change}}{\text{Capacity change}} = \frac{z_G - z_C}{9 - 8}.$$
- $$\frac{\$142 - \$128}{9 - 8} = \$14 \text{ \textbackslash hour}$$
- The point **G** should stays between **B** and **F**.
- The dual price for machine 2 capacity is \$2/hr.



Dual Prices

- The **dual price** is the rate of change of the objective function per unit change of a resource.
- The abstract name "dual" or "shadow" price is standard in LP literature and software packages.
- The dual price of \$14/hr remains valid for changes in machine 1 capacity that move its constraint parallel to itself to any point on the line segment BF .
- The dual price is only valid in the **feasibility range** ($2.67 \text{ hr} \leq \text{Machine 1 capacity} \leq 16 \text{ hr}$), as calculated at points B and F .
- Changes outside this range produce a different dual price (worth per unit).

Changes in the RHS questions

- **Question 1:** If JOBCO can increase the capacity of both machines, which machine should receive priority?

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- **Question 1:** If JOBCO can increase the capacity of both machines, which machine should receive priority?
- **Response:** Priority should be given to machine 1, as each additional hour of machine 1 increases revenue by \$14, as opposed to only \$2 for machine 2.

Changes in the RHS questions

- **Question 1:** If JOBCO can increase the capacity of both machines, which machine should receive priority?
- **Response:** Priority should be given to machine 1, as each additional hour of machine 1 increases revenue by \$14, as opposed to only \$2 for machine 2.
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- **Response:** Only machine 1 should be considered for capacity increase, as the additional net revenue per hour is $14 - 10 = \$4$, compared to a net of $2 - 10 = \$-8$ for machine 2.

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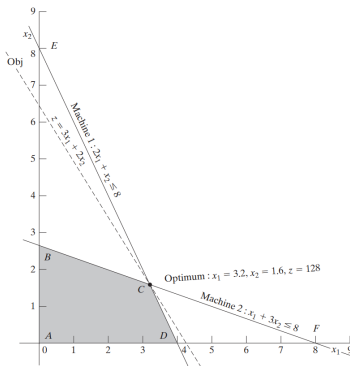
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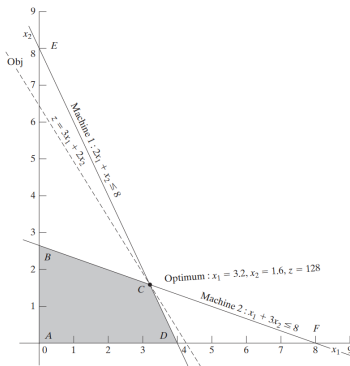
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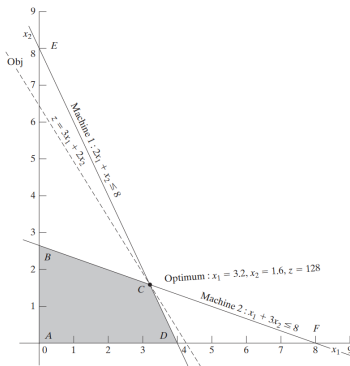
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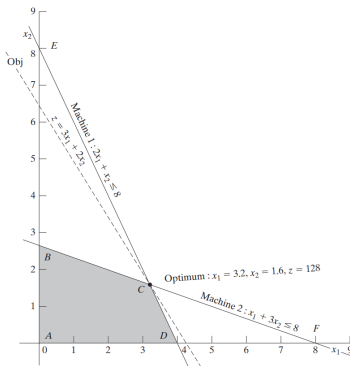
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- **Optimality range** for coefficients keeping optimum at C : $\frac{1}{3} \leq \frac{c_1}{c_2} \leq \frac{2}{1}$.



Objective Coefficient Change questions

- **Question 1:** If unit revenues for Products 1 and 2 are changed to \$35 and \$25, respectively, will the current optimum remain the same?
- The solution at C will remain optimal because $\frac{c_1}{c_2} = \frac{35}{25} = 1.4$ remains within the optimality range $(\frac{1}{3}, 2)$.
- **Question 2:** If the unit revenue of Product 2 is fixed at its current value $c_2 = \$20$, what is the associated optimality range for the unit revenue for Product 1, c_1 , that will keep the optimum unchanged?
- The optimality range for c_1 is: $20 \times \frac{1}{3} \leq c_1 \leq 2 \times 20$.

Conclusion

To sum up, we have covered the following key points:

- The two-phase method provides a viable approach for finding an initial feasible solution for the Simplex method.
- While executing Simplex, one must consider its numerous special cases such as degeneracy, unboundedness, and infeasibility to prevent potential issues.
- Cycling may occur in LP problems with degeneracy, which requires attention to ensure convergence to an optimal solution.
- Graphical sensitivity analysis is a useful tool for investigating the impact of LP parameter changes on the optimal solution, particularly in two dimensions.